AI Chat Manager — Project Concept Document

Reference Document. This is the master concept document for the AI Chat Manager project. You will reference this document throughout all three courses in the AI Development track. Bookmark this page. You will return to it often.

Project Overview

Product Name	**AI Chat Manager**
Project Type	Three-course progressive build (Command-line → Web → AI-integrated)
Status	Concept / Design Phase

Problem Statement

Users of AI assistants (ChatGPT, Claude, Gemini, and others) accumulate hundreds of conversations over time. These conversations contain valuable information: solutions to technical problems, explanations of complex concepts, drafts of documents, and records of decisions. Currently, this knowledge is scattered across platforms, poorly organized, and difficult to search. Users cannot easily find past conversations, extract insights from them, or connect related discussions.

Proposed Solution

A **Personal AI Chat Manager** that allows users to import, organize, search, and analyze their AI conversations. The system will automatically categorize imported chats, enable powerful search across all conversations, and provide an AI assistant that can answer questions using the user's conversation history as context.

Target Users

- **Professionals** who use AI assistants for work and need to retrieve past solutions
- **Students** who use AI for learning and want to review explanations
- **Researchers** who accumulate AI-generated insights and need to organize them
- **Anyone** who wants to transform scattered AI conversations into organized knowledge

Core Capabilities

The finished system will provide the following five capabilities. Click each one to see detailed feature descriptions.

Click each capability to expand its details.

▶ 1. Chat Import and Organization

▶ 2. Search and Retrieval

▶ 3. AI-Powered Analysis

▶ 4. Automation and Agents

▶ 5. User Interface

Technical Architecture

The system will be built in three phases, each corresponding to a course in the program. Each phase builds directly on the previous one.

Click each phase to expand its details.

▶ Phase 1: Command-Line Application (Course 1)

Phase 2: Web Application (Course 2) ▶

Phase 3: AI Integration (Course 3) ▶

Success Criteria

The project will be considered successful when all of the following criteria are met:

1. Users can import chat exports and have them automatically organized
2. Users can search across all conversations and find relevant results
3. Users can ask natural language questions and receive accurate answers based on their conversation history
4. The system runs reliably in a production environment
5. The codebase is well-organized, documented, and maintainable

> **Note:** These criteria apply to the complete system after all three courses. Each course has its own success criteria focused on the phase being built in that course.

AI Concept Mapping

The following table maps the key AI concepts introduced in the program to where they are implemented in this project:

Concept	Course	Where It Is Used
Large Language Models	Course 3	Powering the conversational query interface
Agents	Course 3	Monitoring folders and processing new files automatically

Concept	Course	Where It Is Used
RAG	Course 3	Enabling search across all stored conversations
Fine-Tuning	Course 3	Specialized models for classification tasks
Containerization	Course 2	Docker packaging and Kubernetes deployment

Progression Summary

Course	Output	Key Skills
Course 1	Command-line application	Python, Git, basic ML
Course 2	Web application	Flask, databases, Docker, Kubernetes
Course 3	AI-powered system	Agents, RAG, fine-tuning, automation

> **Each course builds directly on the previous one.** You will extend the same codebase throughout, adding capabilities and sophistication at each stage. This mirrors professional development, where applications evolve over time rather than being built once and abandoned.